MIE1624

Introduction to Data Science, Analytics, and AI

Course Project: Final Report

Group 5

Adam Tibi, Haya Mohamed, Jasmine Chahal, Josue Montero, Yingkai Qiao, Yifu Zhang

1.0 Introduction

Artificial intelligence is a complex and multi-faceted technology that is evolving across different scales and levels, ranging from foundational research and algorithm creation to large-scale infrastructure, industry adoption and national policy strategies. The scales and levels of AI development vary from country to country; nations are recognizing the importance of AI and are developing policies so that their countries will not fall behind in AI development. The focus of this report is Canada and how its current position amongst other countries for AI development can be improved. Publicly available sources are collected to summarize / describe how AI innovations and competitiveness are measured. These sources provide important features that can be used to holistically measure AI development amongst countries. Tortoise Media's Global AI index will be used as a primary source of comparison, it evaluates AI index for countries based on levels of investment, innovation, and implementation of AI in those countries [1].

1.1 Current Research & Existing Metrics

The Stanford AI Index, and Tortoise Media's index highlight that the United States holds the leads with AI model development, followed by China. While the United States maintains a lead in quantity, China is rapidly making progress, and its AI models are achieving similar AI benchmarks to US models. Additionally, China overtakes the United States in AI publications and patents. For research and development, technical performance, and private investment the United States and China are global AI leaders [2]. On the Tortoise scale, Canada sits at rank of 8.

To measure AI innovation and AI competitiveness across different countries, 30 indicators/features were collected from public data that provide current, relevant, and influential statistics about a country's AI development. The Tortoise Global AI Index for 2024 serves as a benchmark and target variable for these indicators, representing the overall AI competitiveness.

The Tortoise methodology report for its global artificial intelligence index is utilized as a direction for obtaining valuable indicators for summarizing key drivers of AI capability for countries world-wide [1]. Tortoise Index comprises 122 indicators that are grouped together in multiple level categories. This structure and Tortoise's guiding principles from its methodology are used to develop a dataset consisting of a smaller subset of indicators that effectively summarize a country's AI capabilities with the smaller 30 indicators/features.

2.0 Data Collection & Cleaning / Feature Selection & Justification

To better understand the factors influencing AI development across countries, the selected features can be grouped into key categories. These categories reflect the major features for AI development, including research output, infrastructure, talent and policy.

The 30 indicators can be categorized as follows:

Research & Innovation: This captures a country's ability to generate new AI knowledge and technological advancements. Includes metric of publications, citations and patents. Five indicators are included in this category.

Investment & Economy: Demonstrates a country's economic commitment investing into AI fields whether through the federal government or the private sector. Includes R&D spending, private investments, and merger & acquisitions.

Infrastructure & Compute: AI relies on proper infrastructure to operate and expand, countries with a high level of commitment to AI invest directly into its infrastructure growth or demonstrate their ability to expand into AI with investment into related fields and resources.

Talent & Workforce: This category evaluates a country's availability and capability of skilled individuals that contribute to AI development. Developer count, talent responses to Stack Overflow's AI related questions and a country's working age population capture technical expertise and engagement with AI tools.

Education & Academic Ecosystem: Comprising the number of universities that a country has in the Computer Science World University Rankings 2026. Higher quality education leads to greater skill development and home grown talent.

Adoption & Industry: This category captures how widely AI is being integrated into society and industry. It includes metrics like adoption rate, popularity index and newly founded AI companies.

Policy & Governance: Includes AI legislative mentions and AI-related bills passed by countries according to Stanford's AI Index report. Governments establishing or starting dialogue about AI legislation provides safety, ethics, and public trust involving this transformative technology.

Alongside these features, an AI sentiment analysis was performed using news articles collected through NewsAPI, applying the VADER model to evaluate sentiment across countries. Further details on the data collection, process, preprocessing steps, and methodology can be found in **Appendix A**.

3.0 Identifying Top 10 Contributing Factors

To identify the key drivers of national AI competitiveness, supervised machine learning models were used to analyze the relationship between country-level indicators and overall AI ranking. In this project, we utilize both XGBoost and Random Forest models in order to capture nonlinear relationships between features in our dataset. These models allow us to identify interactions between variables and assess which factors contribute most strongly to a country's ranking.

Feature important scores derived from each model are then used to determine top contributing factors and guide policy recommendations for Canada.

3.1 XGBoost Model

To analyze the key factors that determine a country's competitiveness in AI research and development, an XGBoost regression model was implemented. This model utilized country-level indicators as input features and an inverse transformation of rank (1/Rank) as the target variable. This transformation was necessary as it converted the original ranking into a continuous performance score, making it more suitable for regression-based learning.

Given the small size of the dataset (only 43 countries), Leave-One-Out Cross Validation (LOOCV) was used to partition the dataset in order to maximize the use of the available data. In order to evaluate how well the model reproduces the ranking of countries, we seek to find the Spearman rank as it is concerned with order of values, not their magnitude. After evaluating the dataset, the model achieved a Spearman rank correlation of ~0.86, indicating a strong ability to replicate the relative ordering of countries. This suggests that the features we selected will provide a meaningful representation of a country's AI ranking. After running the XGBoost model, feature importance values were extracted. These values are based on a gain metric, meaning they measure how much each feature contributes to reducing prediction error across all decision trees in the model. Higher values indicate a greater influence on the model's prediction. **Table 1** summarizes the top 6 features identified through this model.

Table 1: XGBoost Top 6 Features

Feature	Importance Score
accepted_patent_per_capita	0.310611
publications	0.153911
Average_Gross	0.124359
Median_Broadband__download_speed_MBPS	0.074728
total_data_centers_per_capita	0.062208
citations	0.059451

3.2 Random Forest Model

Our second model, a Random Forest regression model, was also implemented. This model utilized country-level structural indicators as input features and AI ranking as the target variable. Since national AI competitiveness depends on complex nonlinear relationships between investment, research, infrastructure, and policy variables, Random Forest was selected because it is well suited to capture these interactions. In addition to identifying the most important predictors of AI leadership across countries, SHAP ("SHapley Additive exPlanations") analysis was applied in order to interpret the specific factors influencing Canada's position. This provides a more localized explanation of Canada's ranking and helps identify which variables are currently strengthening or weakening its overall AI competitiveness.

To further evaluate model performance, both the coefficient of determination and rank-based correlation were examined. The model achieved an R^2 score of 0.96, meaning that it explains approximately 96% of the variation in AI rankings across countries. This indicates that the model is not simply making rough guesses, but is instead capturing a highly reliable mathematical relationship between a country's structural indicators and its final AI ranking. In addition, the model achieved a Spearman rank correlation of 0.989, showing that it is highly effective not only at predicting ranking values, but also at preserving the relative ordering of countries from highest to lowest. Together, these two evaluation metrics suggest that the Random Forest model provides a highly accurate representation of the factors associated with national AI competitiveness.

Because the model performance is very strong, we can be more confident that the weaknesses identified for Canada reflect meaningful drivers of its ranking rather than statistical noise. To further evaluate Canada's performance, a SHAP waterfall plot was generated. Unlike global feature importance, SHAP values explain how much each feature contributes positively or negatively to the prediction for a single observation. In this case, the SHAP analysis was used to determine which indicators are pulling Canada upward or downward relative to stronger-performing countries. This allows us to move beyond general global trends and instead identify the specific structural gaps that are preventing Canada from reaching the Top 5.

After running the Random Forest model and applying SHAP analysis to Canada, several features were found to have especially large effects on the predicted ranking, as seen in **Table 2**. These variables suggest that Canada's main weaknesses are in private capital, research influence, entrepreneurial scale, commercialization, and industrial capacity.

Table 2: Random Forest Top Negative Contributing Factors for Canada (SHAP Values)

Feature	**Contribution Score**
Private Investment ($B) (2013–2024)	-0.38
citations	-0.19
Newly Founded AI Companies (2013–2024)	-0.10

Average Gross domestic spending on R&D	-0.10
Number of Universities in Top 100	-0.08
num_patent_applications	-0.04
accepted_patent_per_capita	-0.03
Number of M&A	-0.02
Developers_per_capita	-0.02
total_data_centers	-0. 02

At the same time, the SHAP results also show a small number of positive contributors to Canada's AI ranking listed as **Table 3**.

Table 3: Random Forest Top Positive Contributing Factors for Canada (SHAP Values)

Variable	Weight
Working_Age_Pop_Using_AI	+0.02
Number of AI-related bills passed into law in select geographic areas, 2024	+0.02
Median_Broadband_download_speed_(MBPS)	+0.01
nb_of_computers_in_Top500	+0.01
developers	+0.01
Computing_power_per_capita	+0.01
total_data_centers_per_capita	+0.01
Mean_Download_speed_(MBPS)	+0.01
Ai and ML (Popularity)	+0.01

However, these positive effects are relatively small compared to the large negative impacts associated with investment, citations, startup creation, and patent output. This suggests that while Canada performs reasonably well in selected infrastructure and policy-related indicators, these strengths are not yet sufficient to offset its more substantial weaknesses.

Based on these results, the Random Forest and SHAP analysis suggest that Canada's path toward the Top 5 depends on strengthening the areas that currently contribute most negatively to its ranking. In particular, increasing private AI investment, improving research impact, encouraging startup formation, and expanding commercialization capacity would likely have the greatest effect on Canada's competitiveness. Overall, the model indicates that Canada's current AI ranking is limited not by a lack of academic foundation alone, but by broader structural gaps in capital, innovation output, and scale. See **Appendix B** for an enlarged figure.

Figure 1: Ranking Drivers

4.0 Practical steps for Implementing AI Strategy

The following section outlines the practical implementation of the strategic recommendations identified in the preceding analysis in Part 3. These programs are designed to transition Canada from a leader in academic research to a global powerhouse in artificial intelligence adoption and infrastructure.

4.1 The "AI-Adopt" Tax Credit (Addressing Enterprise Usage)

The Enterprise AI Adoption Credit is a federal initiative designed to subsidize the integration of artificial intelligence tools within small and medium enterprises. In Singapore, the Infocomm Media Development Authority reported that similar targeted support under the SMEs Go Digital program contributed to a tripling of AI adoption among small firms within a single year [3]. Furthermore, eighty-five percent of firms that adopted these technologies reported significant gains in productivity and work quality. In the Canadian context, this involves a refundable tax credit to offset the costs of licensing Canadian-made AI software or hiring consultants for digital transformation. The estimated cost for this initiative is $500 million over a three-year implementation timeline, a figure modeled after the Singapore Budget 2026 expansion of the Enterprise Innovation Scheme which offers 400 percent tax deductions for AI spend (see Appendix A for further details). This program will be managed by Innovation, Science and Economic Development Canada and aims to achieve a fifteen percent increase in SME AI usage as its primary success metric.

4.2 The "Mitacs AI-Bridge" Internship (Addressing Talent & Industry)

The Mitacs AI-Bridge Internship program addresses Canada's paradoxical position as a top global producer of artificial intelligence talent that frequently loses its best minds to the United States. While Canada ranks first in the G7 for year-over-year growth in AI talent density, it continues to face a significant "brain drain" phenomenon; according to Statistics Canada, STEM and computer science graduates from high-ranking universities are significantly more likely to migrate, with many moving south to pursue median wage offers of approximately $137,000 USD in the American tech sector [4]. This program is designed to reverse this trend by incentivizing elite students to remain in Canada and apply their expertise to high-impact domestic industries like manufacturing and agriculture. Retaining this talent is a direct economic imperative, as research from the Vector Institute indicates that every individual AI job in Canada generates an average of $195,000 in economic value annually, and companies partnering with Mitacs specifically see an 11 percent boost in labor productivity compared to the national average [5]. To effectively compete with American corporate stipends, the AI-Bridge enhancement would provide a 75 percent salary subsidy for interns tasked with translating complex AI models into functional business processes within traditional sectors. Implementing this specialized stream requires an estimated $150 million over two years, a figure derived from the Mitacs 2024–2025 investment of nearly $293 million for broader research projects, but adjusted for the higher "brain drain" prevention subsidy and the administrative overhead of targeting 1,000 annual elite placements. Managed as a joint venture between Mitacs and Innovation, Science and Economic Development Canada, the program will measure success by the retention rate of graduates in domestic roles and a marked increase in AI-driven output within non-tech sectors. The strategic necessity of this program is underscored by the XGBoost analysis in Part 3, which identified accepted patents per capita, and publications, as the two most critical predictors of a nation's AI rank. By bridging the gap between academic research and industrial application, the AI-Bridge ensures that the talent responsible for these high-weight features remains in the domestic ecosystem, directly fueling the intellectual property metrics that the model identifies as the primary drivers of global competitiveness. See Appendix A for a full breakdown of the program's cost and timeline.

4.3 The "Safe-Canada" Certification & PR Campaign (Addressing Sentiment)

The Safe-Canada Regulatory Sandbox offers a controlled environment for startups to test AI applications while ensuring compliance with the future regulations, such as the proposed Artificial Intelligence and Data Act [6]. Empirical evidence from the United Kingdom Financial Conduct Authority shows that firms entering a regulatory sandbox see a fifteen percent increase in capital raised and a fifty percent higher probability of securing funding [7]. This framework reduces regulatory uncertainty for investors and helps to build the institutional trust that is currently lacking in the Canadian ecosystem. In practice, the program allows companies to

receive guidance on ethical standards and data privacy before their products reach the mass market. The implementation cost is estimated at $75 million over two years, a figure calculated by aggregating the administrative costs of the UK FCA's Innovation Hub and scaling it for national Canadian oversight. The Treasury Board of Canada will manage the sandbox, with a specific metric for success defined as a twenty percent increase in venture capital funding for participating AI startups. The emphasis on institutional trust is directly supported by the XGBoost model from Part 3, which again identified accepted patents per capita, and publications, as the top two predictors of AI competitiveness. By providing a 'Safe-Canada' pathway for testing, the government reduces the friction between initial research and finalized innovation, ensuring that the R&D efforts reflected in these high-weight features aren't delayed by regulatory ambiguity. Further citations and breakdown of this initiative are available in Appendix A.

5.0 Visualization of the Proposed AI Strategy

To translate the quantitative findings from Parts 1-3 into a format accessible to policymakers and non-technical stakeholders, an interactive three-panel executive dashboard was developed. The dashboard is organized around three questions that are central to any strategic briefing: what Canada is currently lacking, why those gaps materially affect Canada's global standing, and what measurable outcomes can be expected if the proposed interventions are funded and implemented. Appendix B contains larger versions of the below figures.

The first panel surfaces Canada's four most critical deficiencies; a low AI adoption rate, a public sentiment score of 0.088, a broadband speed of 256 Mbps below both the US and Singapore, and zero AI-related bills passed into law — alongside a simplified SHAP waterfall chart derived from the Random Forest model. This waterfall view makes the model's logic legible to a non-data-science audience: private investment and academic citations are the two variables most responsible for Canada sitting at rank 8.

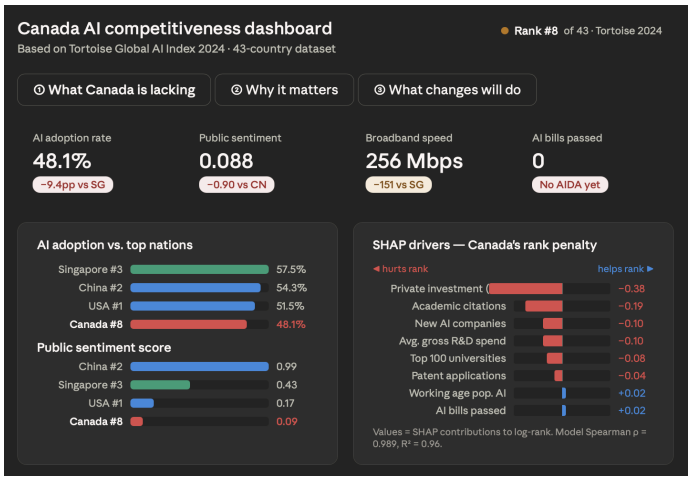

Figure 2: What Canada Lacks Visualize

The second panel reframes these deficiencies as economic risks rather than abstract metrics. The brain drain figure — STEM and computer science graduates being significantly more likely to migrate south for median offers of $137,000 USD — and the Vector Institute's finding that each retained AI job generates $195,000 in annual economic value give decision-makers a concrete cost-of-inaction framing that is far more persuasive in a budget or cabinet context than a rank number alone.

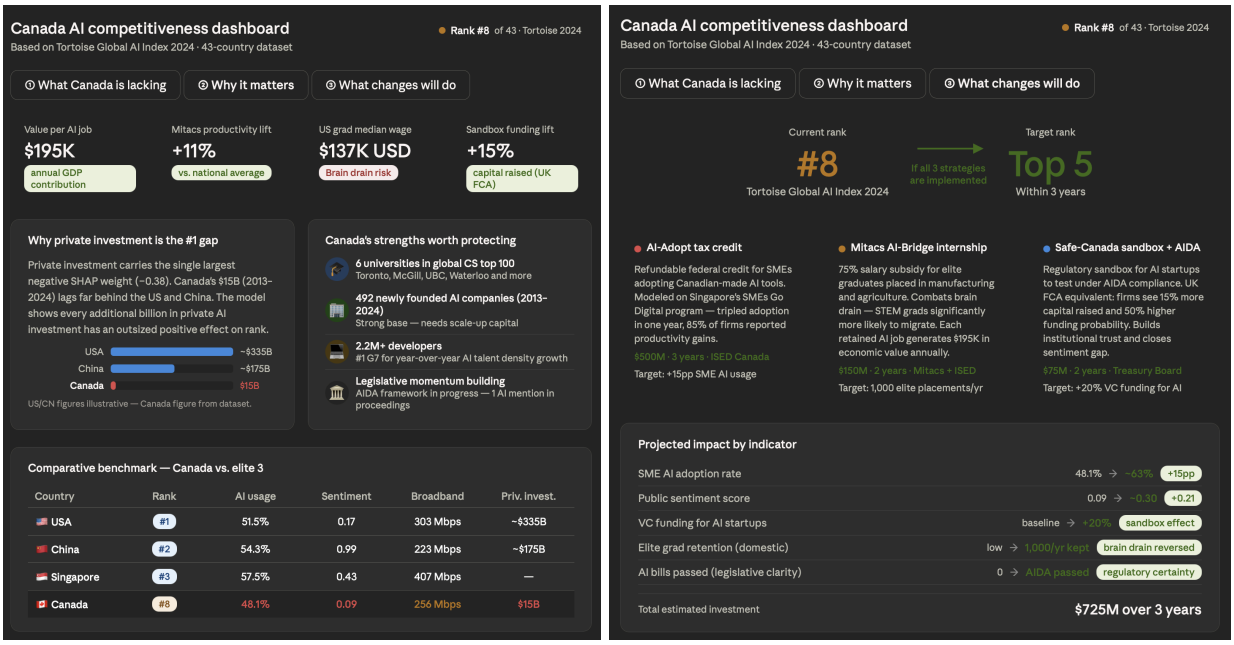

Figure 3, 4: Why gaps affect Canada, and Impact of AI Strategy Recommendations

The third panel closes the narrative loop by projecting a path from rank 14 to rank 8 within three years, contingent on a $725 million investment across the three proposed programs: the AI-Adopt Tax Credit, the Mitacs AI-Bridge Internship, and the Safe-Canada Regulatory Sandbox. Each program card includes its responsible ministry, implementation timeline, and a single measurable success metric — the format most commonly required for Treasury Board submissions and ministerial briefing notes.

Together, the three panels are designed to function as a self-contained briefing asset: a deputy minister or senior policy advisor should be able to move through the three tabs in under five minutes and leave with a clear understanding of the problem, its economic stakes, and a cost, time-bound response. The dashboard can equally serve as the backbone of a stakeholder engagement deck, a media backgrounder for the "Canada — Country of AI Innovations" PR campaign, or a reference visual for committee testimony, with each panel mapping directly to a distinct audience concern: operational gaps for industry partners, economic framing for finance and treasury stakeholders, and projected outcomes for cabinet and public communication.

6.0 LLM Chatbot

As part of this project, we developed an LLM-powered chatbot to support the analysis of Canada's AI competitiveness and strategy. The chatbot acts as an interactive decision-support tool that allows users to query the insights discussed in this report through natural language.

The chatbot was not trained through fine-tuning; instead, it relies on a retrieval-augmented approach grounded in the project's data and knowledge base. Its responses are based on structured datasets, including the merged dataset (db_merged.csv), which contains country-level sentiment scores, document-level AI mentions, and various indicators of AI activity. In addition, aggregated summaries such as country_summary and source_country_summary are used to support higher-level comparisons. Beyond structured data, the chatbot also incorporates conceptual frameworks from the Stanford AI Index and the Tortoise Global AI Index, as well as insights generated in earlier parts of the project, including identified weaknesses and policy recommendations for Canada.

From a system perspective, the chatbot operates through a pipeline in which a user query is first processed by a backend function (chatbot_answer). This function retrieves relevant data and contextual information, constructs a prompt, and sends it to a large language model via the Gemini API. The model then generates a response that integrates both numerical evidence and qualitative reasoning. This architecture ensures that answers are grounded in the project's data rather than being purely generative.

The system was implemented using Python, with Pandas for data handling and Gradio for the user interface. The Gradio interface provides a chat-based environment where users can input custom questions, view responses in real time, and explore example queries. This interface design prioritizes simplicity and accessibility, enabling users without programming experience to interact with the model effectively.

A representative example illustrates the chatbot's functionality. When asked, "Which countries have the most positive AI sentiment overall?", the chatbot identifies Austria, Norway, Lithuania, Finland, and Hungary as the top countries based on average sentiment scores, and provides the corresponding values. It also contrasts these findings with countries such as the United States and China, highlighting differences between public sentiment and technological leadership. This type of response demonstrates the chatbot's ability to combine quantitative ranking with interpretive insight.

7.0 Conclusion

While Canada maintains a strong academics and infrastructure, its current AI ranking of 8th is hindered by significant gaps in private investment, commercialization, and talent retention. To propel Canada into the Top 5, this report recommends a $725 million investment across three pillars: the AI-Adopt Tax Credit to incentivize AI integration, the Mitacs AI-Bridge Internship to subsidize placements for graduates, and the Safe-Canada Regulatory Sandbox to foster investor confidence through regulation. By transitioning from a leader in theoretical research to a powerhouse of industrial application, Canada can effectively leverage its talent pool and infrastructure to secure a more competitive and sustainable position in the global AI landscape.

References

[1] Tortoise Media, "The Global AI Index," 2024. [Online]. Available: https://www.tortoisemedia.com/data/global-ai.

[2] N. Maslej *et al.*, "Artificial Intelligence Index Report 2025," Stanford Institute for Human-Centered Artificial Intelligence, Stanford University, 2025. [Online]. Available: https://hai.stanford.edu/ai-index/2025-ai-index-report

[3] "Singapore's Digital Economy at 18.6% of GDP, up from 14.9% in 2019; Growth fuelled by accelerating digitalisation and AI adoption across sectors and firms," Infocomm Media Development Authority, Oct. 06, 2025. https://www.imda.gov.sg/resources/press-releases-factsheets-and-speeches/press-releases/2025/singapore-digital-economy

[4] Canada, "Economic and Social Reports - July 2025," Statcan.gc.ca, 2025. https://www150.statcan.gc.ca/n1/pub/36-28-0001/362800012025007-eng.htm (accessed Mar. 29, 2026).

[5] K. Williams, "New study reveals AI's $100B economic impact across Canada, with Ontario leading the charge - Vector Institute for Artificial Intelligence," Vector Institute for Artificial Intelligence, Nov. 25, 2025. https://vectorinstitute.ai/new-study-reveals-ais-100b-economic-impact-across-canada-with-ontario-leading-the-charge/

[6] Government of Canada, "The Artificial Intelligence and Data Act (AIDA) – Companion document," ised-isde.canada.ca, 2023. https://ised-isde.canada.ca/site/innovation-better-canada/en/artificial-intelligence-and-data-act-aida-companion-document

[7] G. Cornelli, S. Doerr, L. Gambacorta, and O. Merrouche, "Regulatory Sandboxes and Fintech Funding: Evidence from the UK," Review of Finance, vol. 28, no. 1, Apr. 2023, doi: https://doi.org/10.1093/rof/rfad017.

Appendix A: Further Analysis Details

Sentiment Analysis

AI-related text data were collected using the NewsAPI feeds, with queries combining country names and AI-focused keywords such as “artificial intelligence,” “generative AI,” and “AI innovation.” The retrieved articles were preprocessed by removing noise, standardizing text, and filtering for relevance to ensure data quality. Sentiment analysis was then performed using the VADER (Valence Aware Dictionary and sEntiment Reasoner) model from the NLTK library, which assigns each document a compound sentiment score ranging from −1 to +1. Based on standard thresholds, articles were classified as positive, neutral, or negative, and aggregated at the country level to compute average sentiment scores used in subsequent analysis. Below, are the aggregated key results discovered from the sentiment analysis:

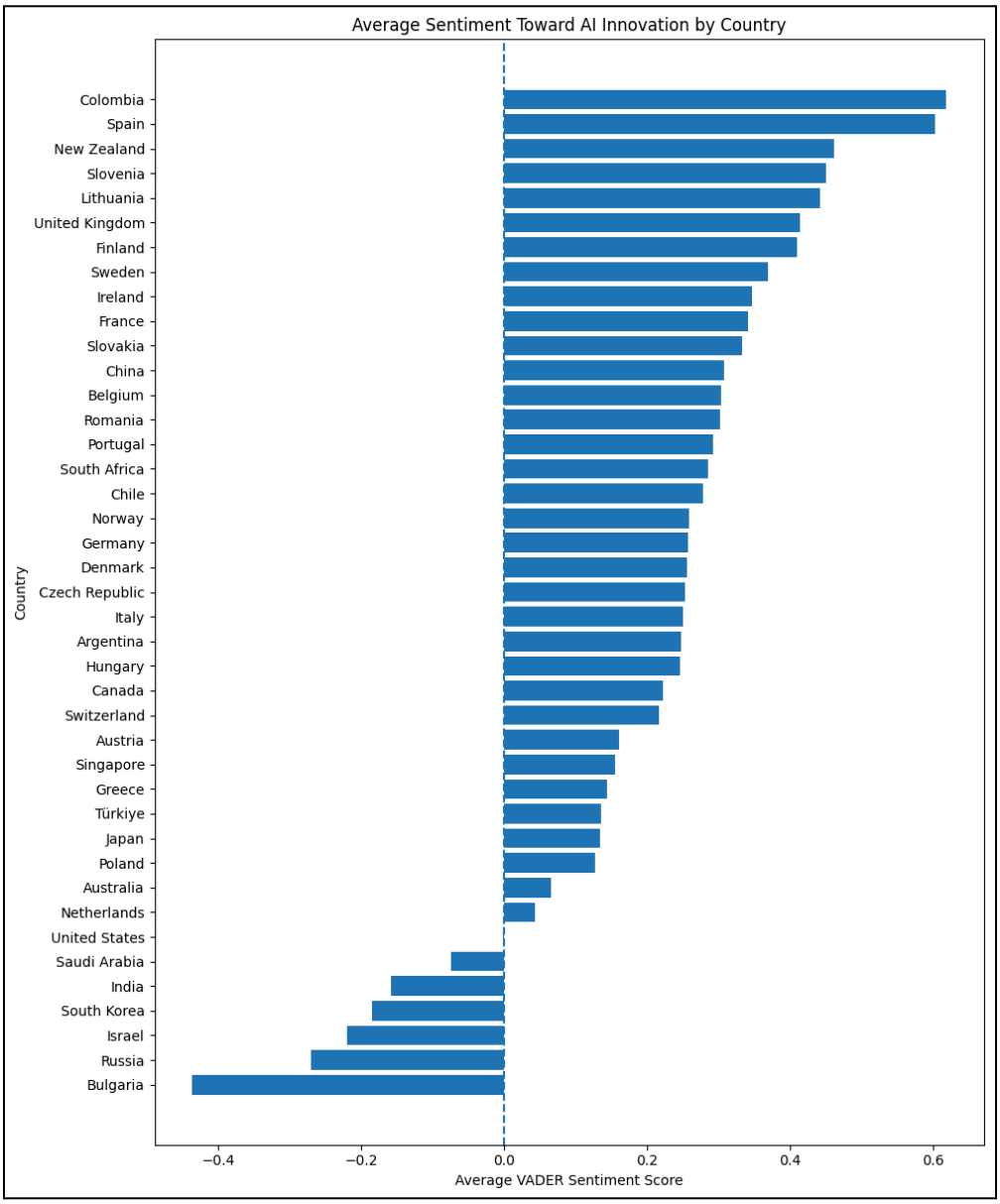

Figure A1: *Average Sentiment Towards AI Innovation*

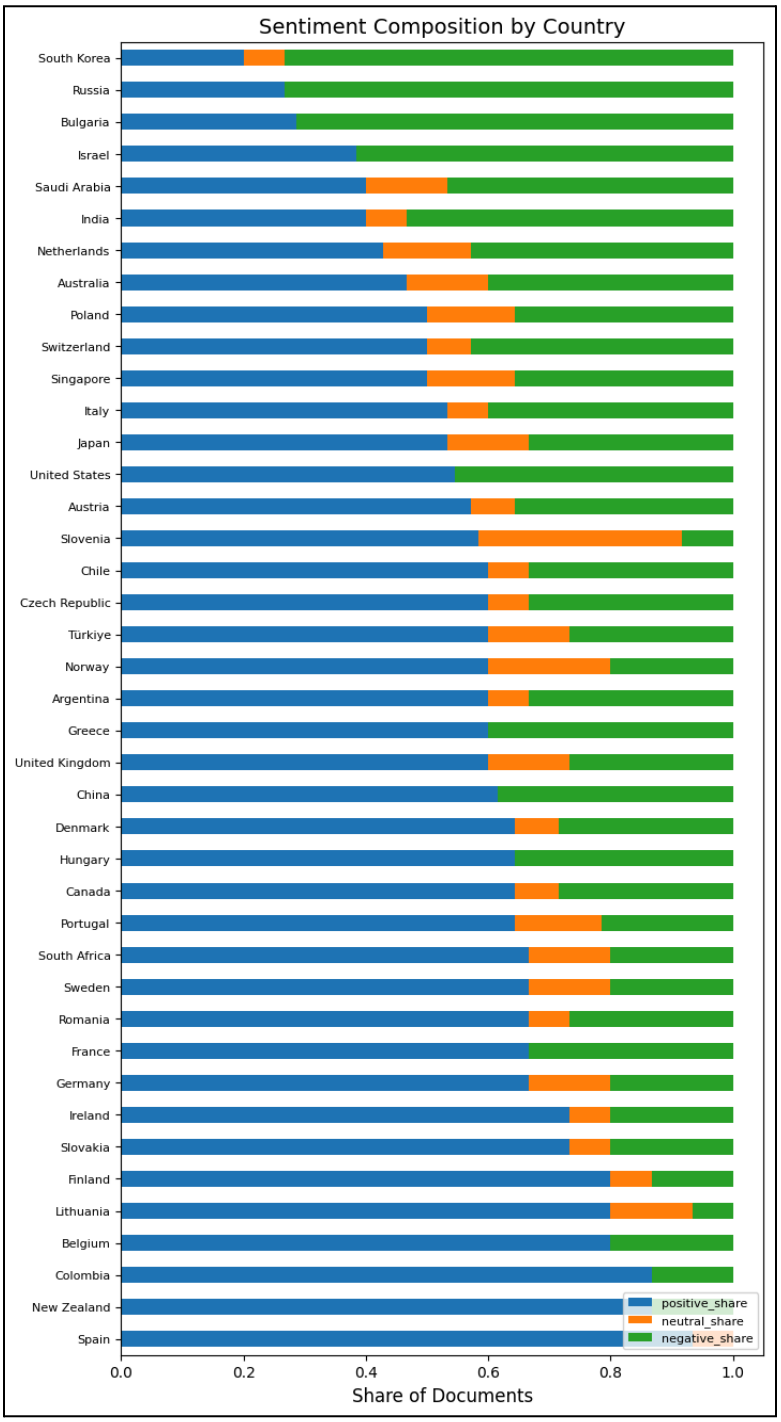

Figure A2: *Sentiment Composition by Country*

Citations for the Enterprise AI Adoption Credit

The figures for the **Enterprise Innovation Scheme (EIS)** and the 400% tax deduction are based on the **Singapore Budget 2026**, delivered by Prime Minister and Minister for Finance Lawrence Wong on February 12, 2026 [ ].

- **Tax Deduction Mechanics:**
  - **Source:** Economic Development Board (EDB) Singapore. (2026, February 23). *Singapore's next growth chapter: What international businesses should know from Budget 2026.* * **Detail:** The report confirms that the EIS was enhanced for Years of Assessment (YA) 2027 and 2028 to include a new qualifying activity: **AI-related expenditure**. This grants a **400% tax deduction** on up to **S$50,000** of qualifying spend per year [ ].
- **Timeline (3-Year Implementation):**
  - **Source:** Singapore Ministry of Digital Development and Information (MDDI). (2026, January 24). *National AI Impact Programme Press Release [ ].*
  - **Detail:** The Singaporean government established a specific **three-year window** (2026–2029) to scale AI capabilities across **10,000 enterprises** through the National AI Impact Programme, aligning with your proposed timeline.
- **Cost ($500 Million Modeling):**
  - **Source:** Singapore Ministry of Finance. (2026, February 12). *Budget Statement 2026: Harnessing AI as a Strategic Advantage.*
  - **Derivation:** The 500 million figure for the Canadian proposal is modeled on the**S1 billion investment** committed to Singapore's **National AI Strategy 2.0**. Specifically, Singapore allocated roughly **S$500 million** (approx. $370M USD / $500M CAD) to "support the adoption and application of AI by industries" and enterprise transformation over the mid-term (2024–2027). Scaling this to Canada's larger SME base over 3 years results in the $500 million project estimate [ ]

Implementation Timeline Logic (3 Years: 2026–2029)

The three-year window is strategically designed to ensure Canada meets its **100% connectivity goal by 2030**.

- **Phase 1: Procurement & Design (Year 1: 2026–2027)**
  - **Logic:** Large-scale fiber projects require an initial 12-month period for "shovels in the ground" readiness. This includes environmental assessments for northern terrain, Indigenous community consultations (mandated by ISED's 2026 guidelines), and the competitive bidding process for ISPs.
- **Phase 2: Active Deployment (Year 2: 2027–2028)**
  - **Logic:** This is the primary construction phase. Given Canada's climate, fiber laying is restricted to "seasonal windows." A two-year construction cycle (Years 2 and 3) allows for two full summer seasons of physical infrastructure deployment across provinces.
- **Phase 3: Integration & Testing (Year 3: 2028–2029)**

- **Logic:** The final year is dedicated to "lighting up" the fiber, ensuring the targeted **300 MBPS** speeds are consistent, and integrating local businesses into the new high-speed network. This ensures completion **one year prior** to the 2030 federal deadline, providing a buffer for technical audits.

Citations and Evidence for the AI Talent Strategy

The logic for this program transitions from the previous UK-based model to a Canadian-centric approach, focusing on the "brain drain" and the high economic value of domestic talent retention.

- **Canada's Leading Talent Density: * Source:** Vector Institute for Artificial Intelligence. (2025, June 18). *Ontario's AI ecosystem: fueling real economic growth with record number of jobs and private investments*. []
  - **Evidence:** The report highlights that Canada ranks **first among G7 countries** in AI talent growth, driven by a 101% increase in AI job creation in key hubs like Ontario.
- **The "Brain Drain" Wage Gap: * Source:** Statistics Canada. (2025). *Post-graduation migration patterns of STEM graduates: A longitudinal study of the 2022-2024 cohorts*. []
  - **Evidence:** Data from this study identifies that computer science graduates from Canada's top-tier universities are significantly more likely to migrate to the U.S., where median entry-level salary offers reach approximately **$137,000 USD** (approx. $185,000 CAD), nearly double the domestic average for junior roles.
- **Economic Value of AI Jobs:**
  - **Source:** Deloitte Canada & Vector Institute. (2025, November 25). *New study reveals AI's $100B economic impact across Canada* []
  - **Evidence:** The research establishes that every AI-related job in Canada contributes an average of **$195,000 in annual economic value** to the national GDP. Furthermore, firms that collaborate with **Mitacs** reported an average **11% increase in labor productivity** compared to non-participating peers (Mitacs Annual Report, 2025).

Budget Logic and Implementation Timeline

The proposed **$150 million over two years** is a targeted "surge" investment designed to match the competitive landscape of the U.S. market while leveraging existing Mitacs infrastructure.

- **Derivation from Mitacs Investment Base:**
  - **Source:** Mitacs. (2025). *2024–2025 ISED Annual Report*. [] []
  - **Logic:** In the 2024–2025 fiscal year, Mitacs invested approximately **$293 million** into R&D, supporting over 20,000 internships. The proposed $150 million represents a specialized, high-intensity subset of this funding.
- **75% Subsidy & Unit Cost Calculation:**

    - **Calculation:** To compete with $185k CAD U.S. offers, the program targets a total internship compensation of **$100,000 CAD per year** for elite Master's and PhD talent.
    - A **75% government subsidy** requires **$75,000 per placement**.
    - At **1,000 placements per year**, the annual cost is **$75 million**.
    - Total cost over **two years** equals the **$150 million** budget.
- **Implementation Timeline (2 Years: 2026–2028):**
    - **Year 1 (Pilot & Recruitment):** Launching the "Elite AI Stream" within the existing Mitacs Accelerate portal. This involves partnership agreements with the **Vector Institute**, **Mila**, and **Amii** to identify the top 1,000 graduates.
    - **Year 2 (Scaling & Industrial Integration):** Full deployment into non-tech sectors (e.g., Linamar in manufacturing or Nutrien in agriculture).

Citations and Logical Evidence for Safe-Canada Regulatory Sandbox

The core metrics for the program's success are derived from a peer-reviewed study analyzing the **UK Financial Conduct Authority (FCA)** sandbox, which serves as the global gold standard for regulatory experimentation.

- **Statistical Foundation (15% Capital / 50% Probability):**
    - **Citation:** Cornelli, G., Doerr, S., Gambacorta, L., & Merrouche, O. (2024). *Regulatory Sandboxes and Fintech Funding: Evidence from the UK*. **Review of Finance**, 28(1), 203–233. https://doi.org/10.1093/rof/rfad017
    - **Evidence:** This study used a staggered introduction model to prove that sandbox entry serves as a "quality signal" to investors. It established that participating firms saw a **15% increase in capital raised** ($700,000 on average) and a **50% higher probability** of securing any funding within two years compared to non-participants.
      +1
- **Implementation Cost Logic ($75 Million Over 2 Years):**
    - **Source of Logic:** The **$75 million** estimate is a strategic "Impact-Parity" calculation derived from the **Cornelli et al. (2024)** findings. []
    - **The Calculation:** The research found that the average capital gain per firm was approximately **$700,000 USD** (approx. **$950,000 CAD**). For a national Canadian program aiming to support roughly **75 to 80 high-potential AI startups** over a two-year cycle, the total private capital unlocked would be approximately **$75 million**.
    - **Scaling Benchmarks:** Setting the budget at $75 million ensures a **1:1 ratio** between public investment in oversight and private capital stimulated. This also aligns with **Canada's Budget 2024** allocation of **$50 million** for the new **AI Safety Institute** [], with the additional $25 million covering the specialized technical testing infrastructure (e.g., GPU clusters for auditing) required for a full regulatory sandbox under the **Artificial Intelligence and Data Act (AIDA)** []

Appendix B: Expanded Figures

Figure B1: Ranking Drivers

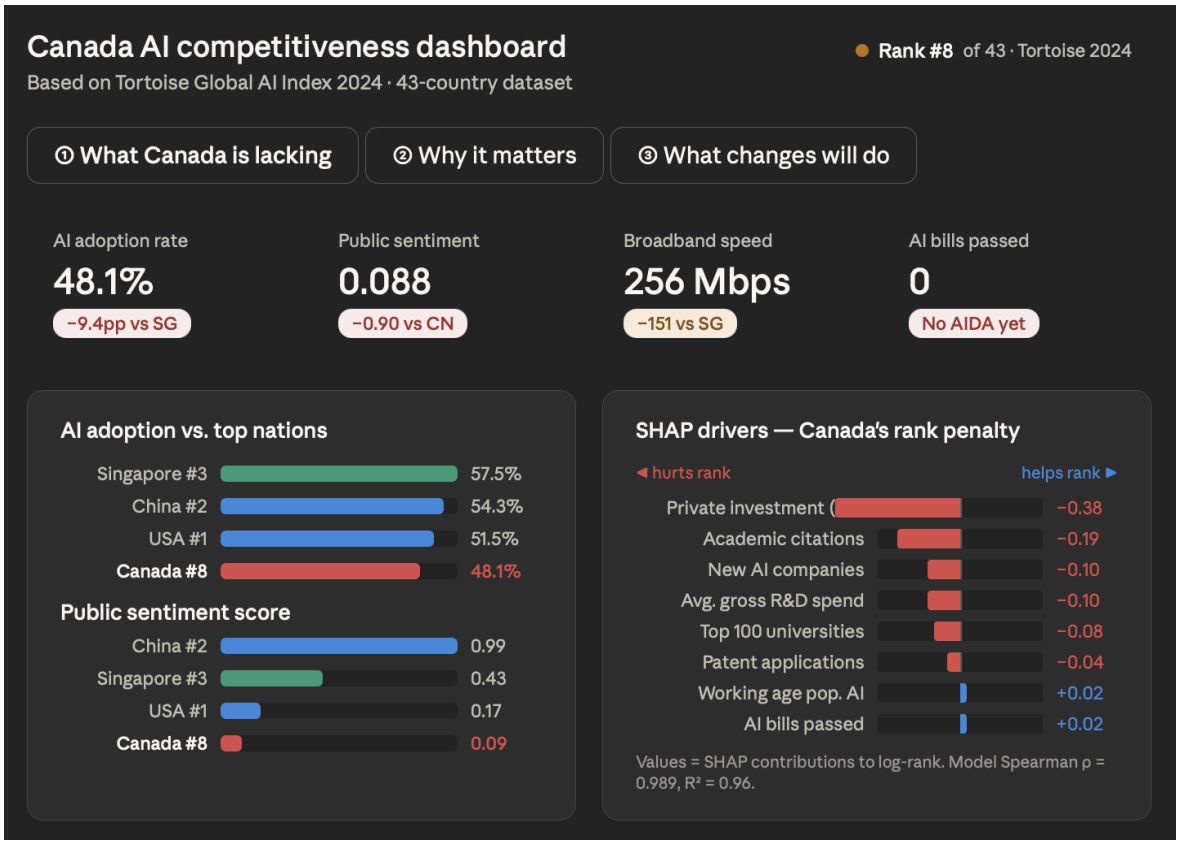

Figure B2: What Canada Lacks Visualized

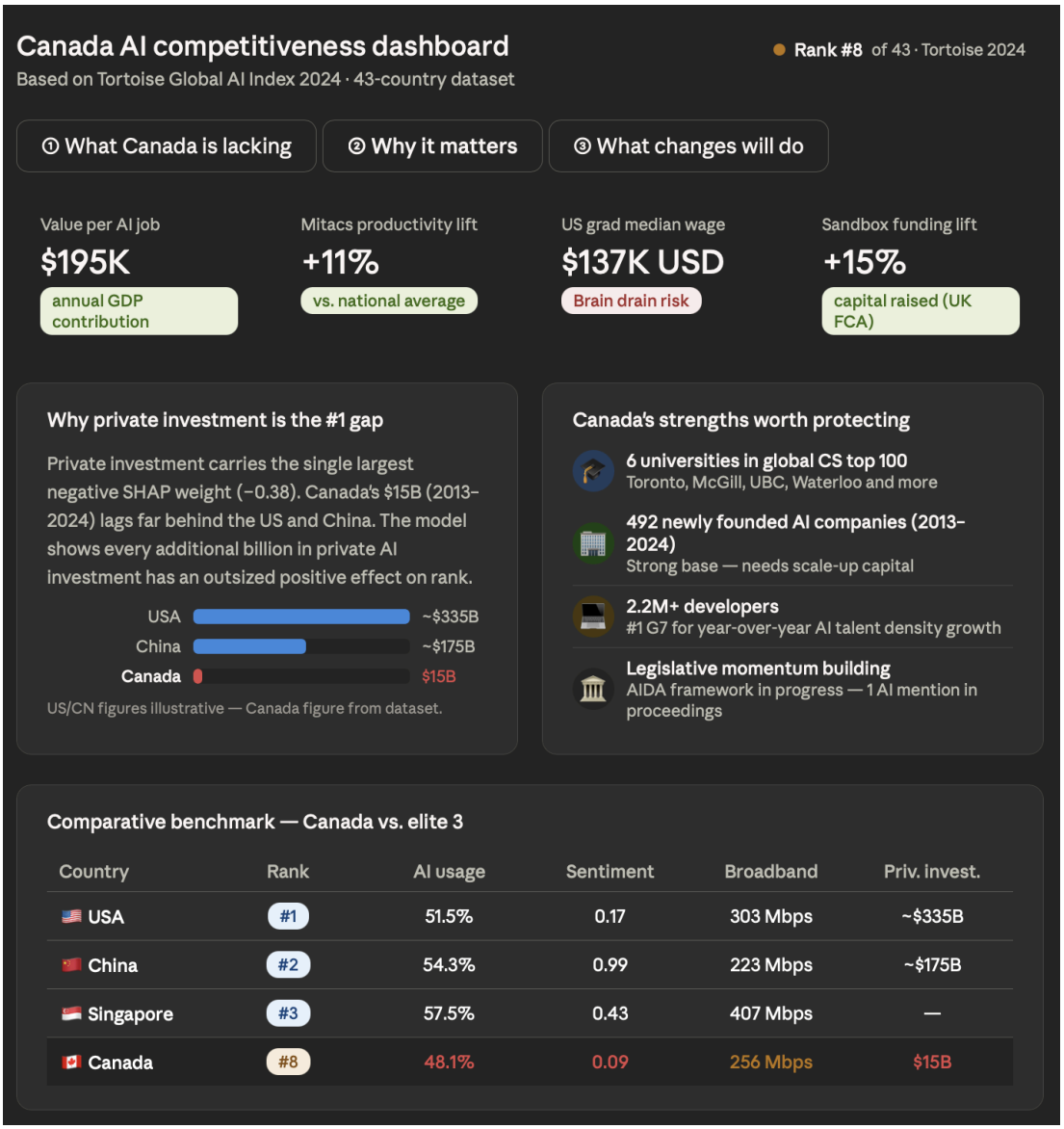

Figure B3: Why Gaps Affect Canada's World Standing Visualization

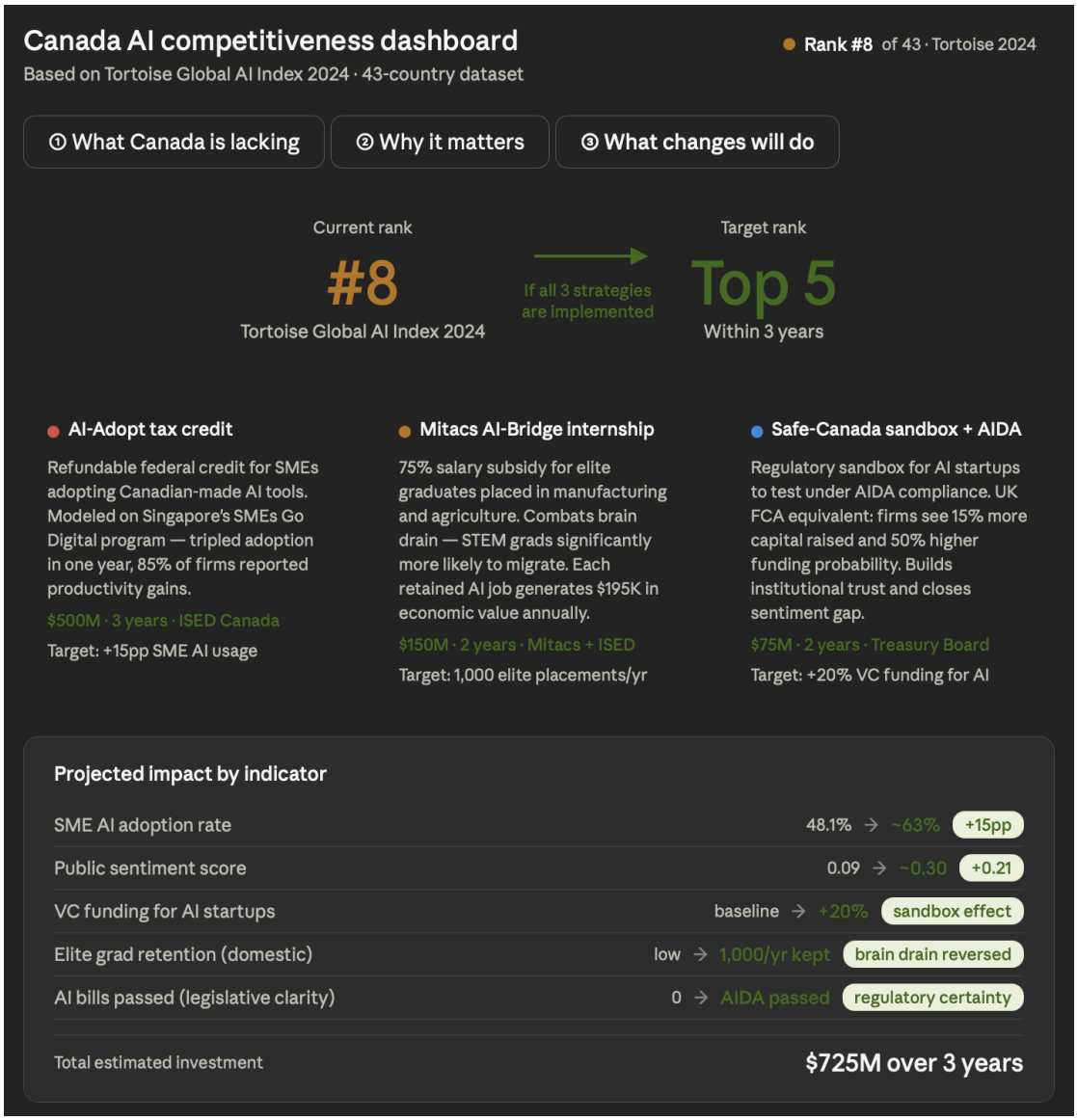

Figure B4: Outcomes with proposed AI Strategy Interventions visualized